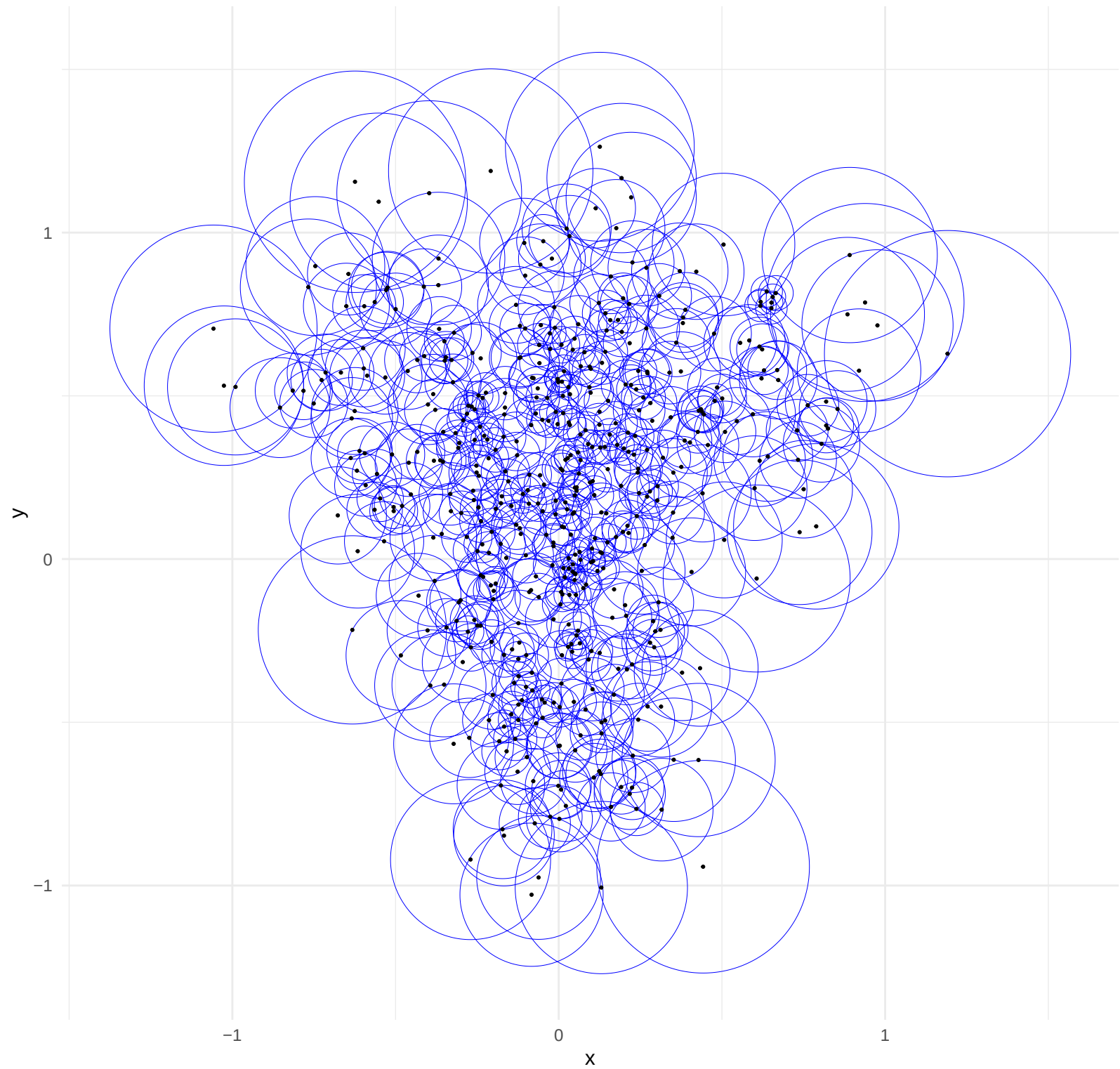
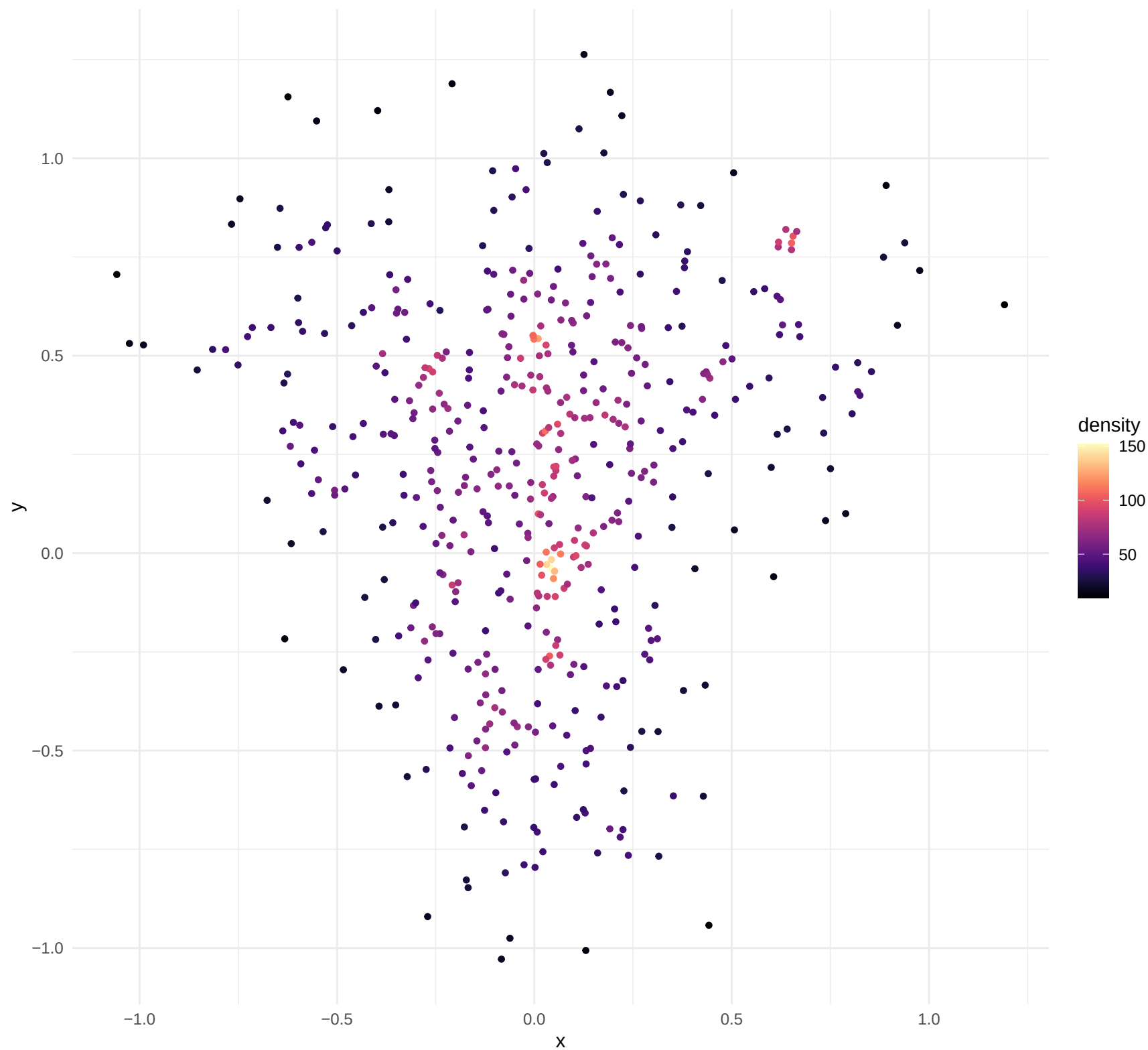


# All Adaptive Spheres

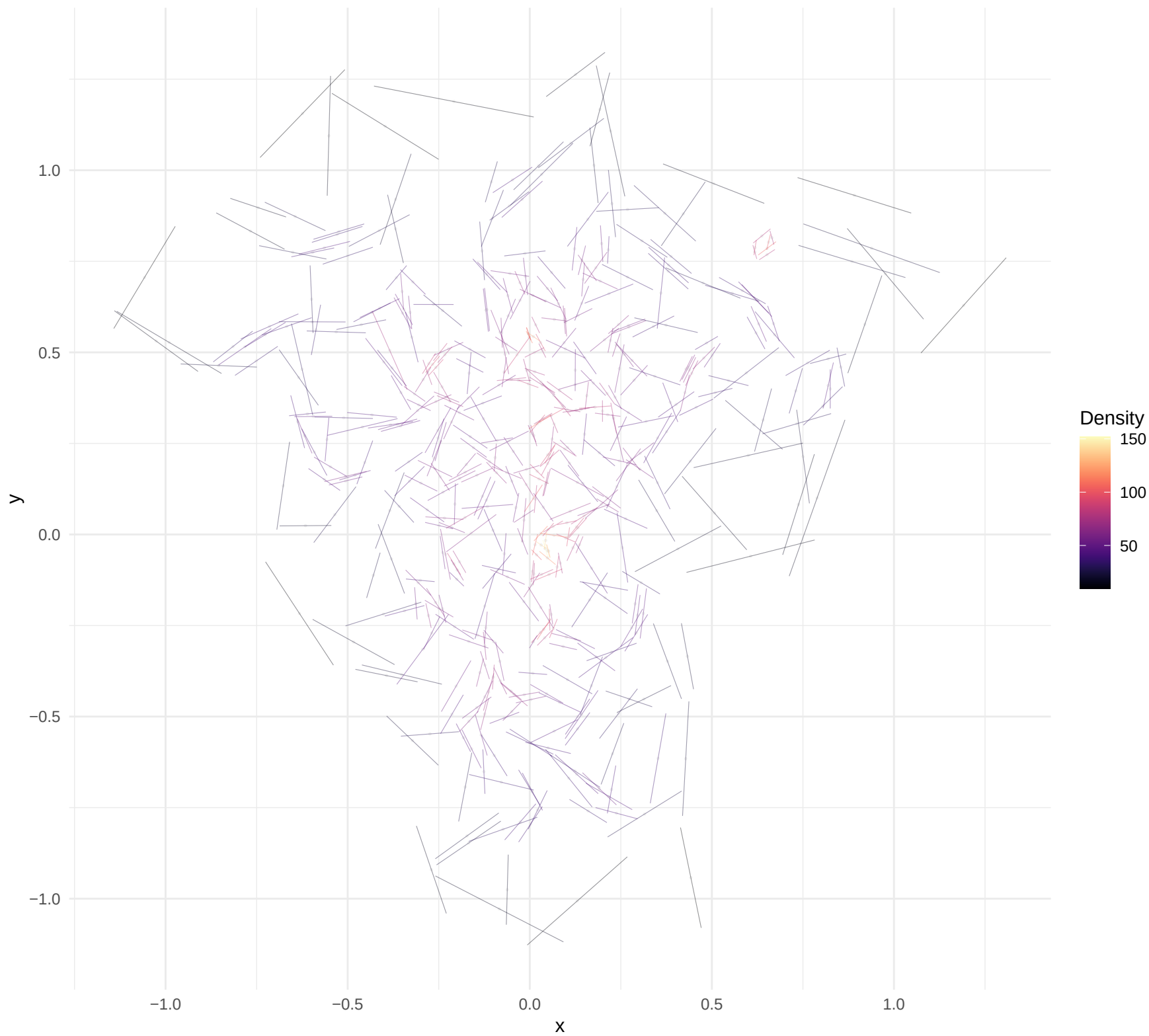
Visualizing local neighborhood size (radius) for every point.



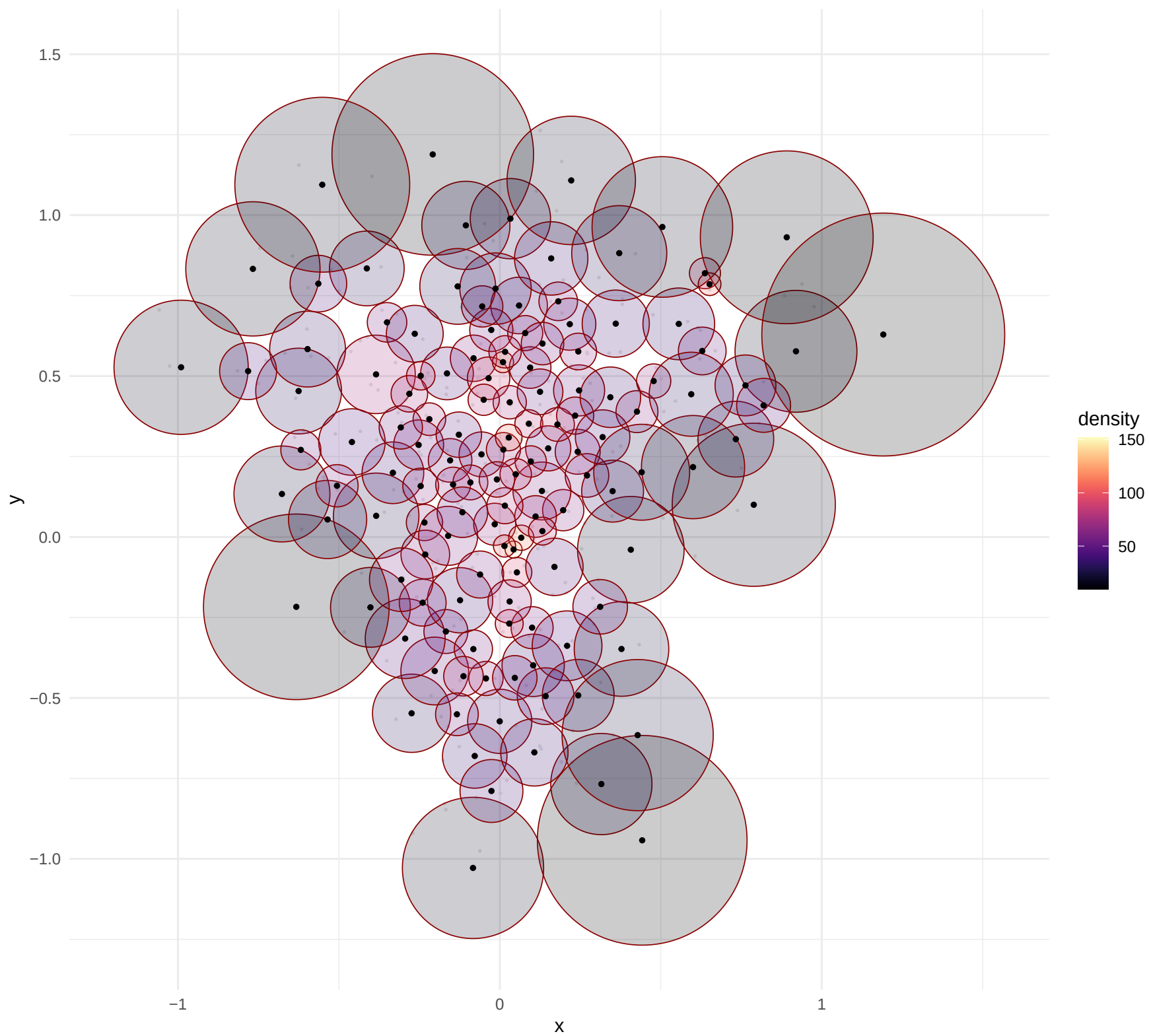
Point Density Distribution



Global Tangent Field (PC1)

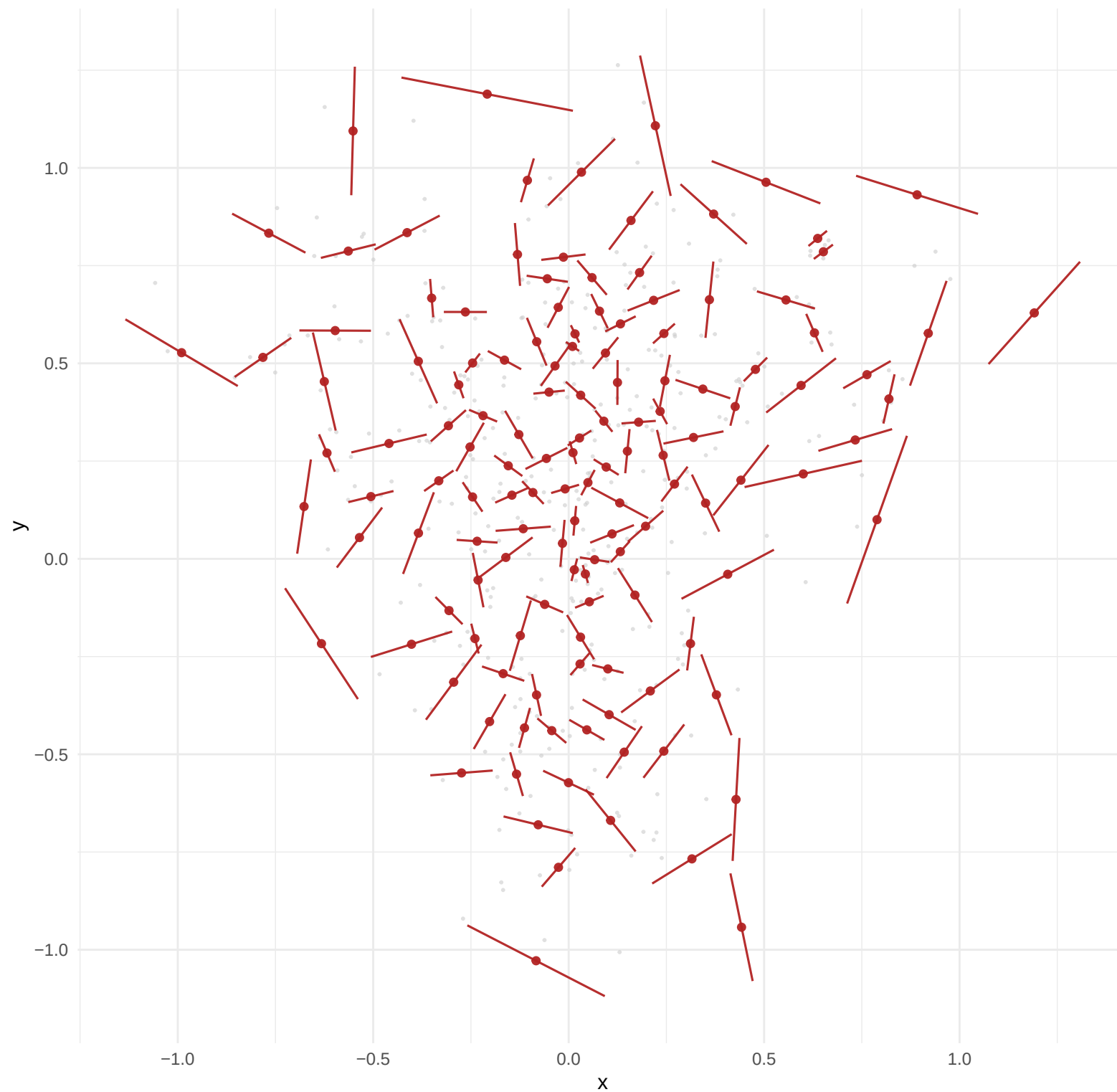


Atlas Selection (Anchors)



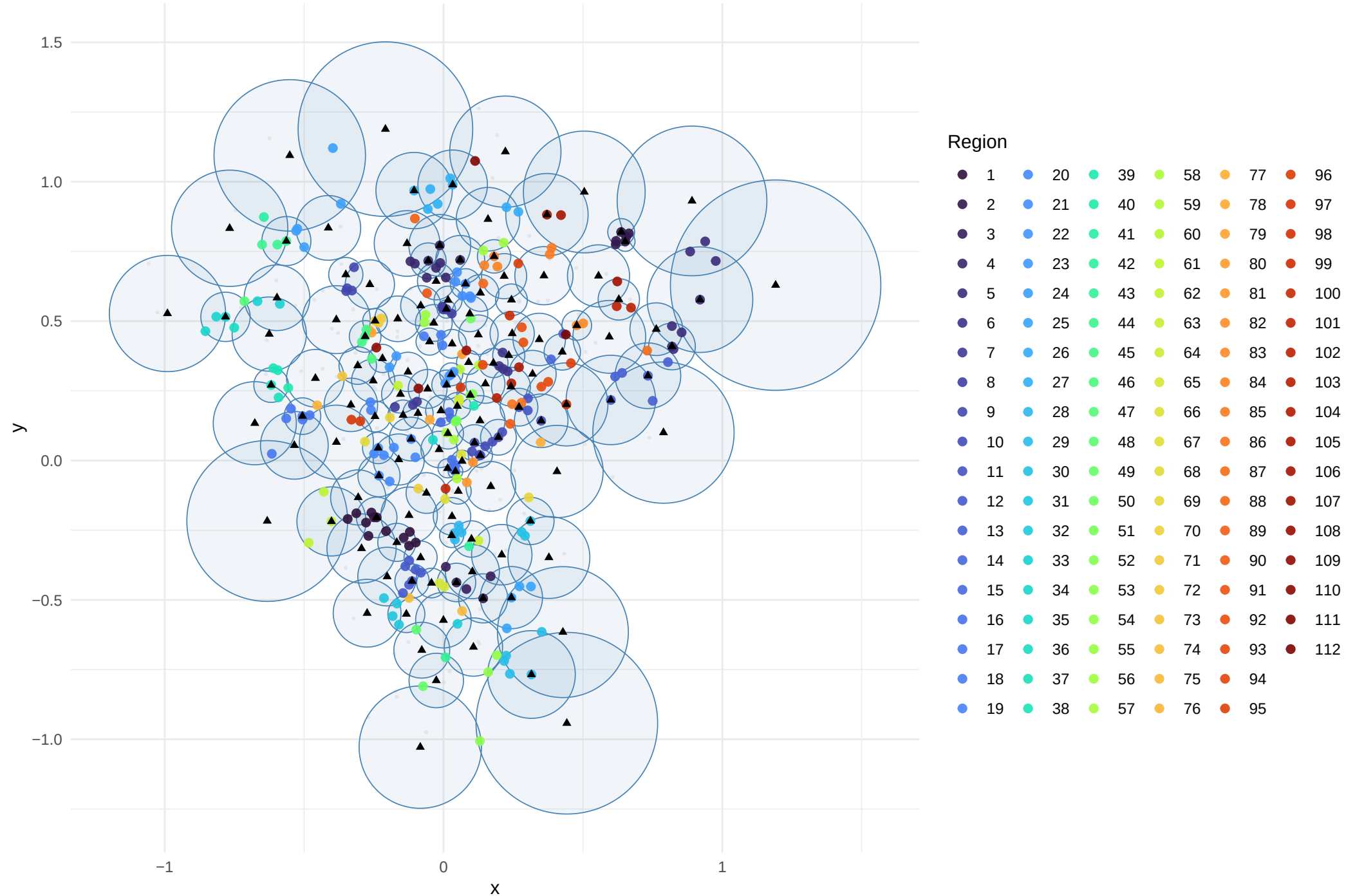
# Anchor SVD Segments

Same anchors as plot 4: spheres first, then their SVD (PC1) in red



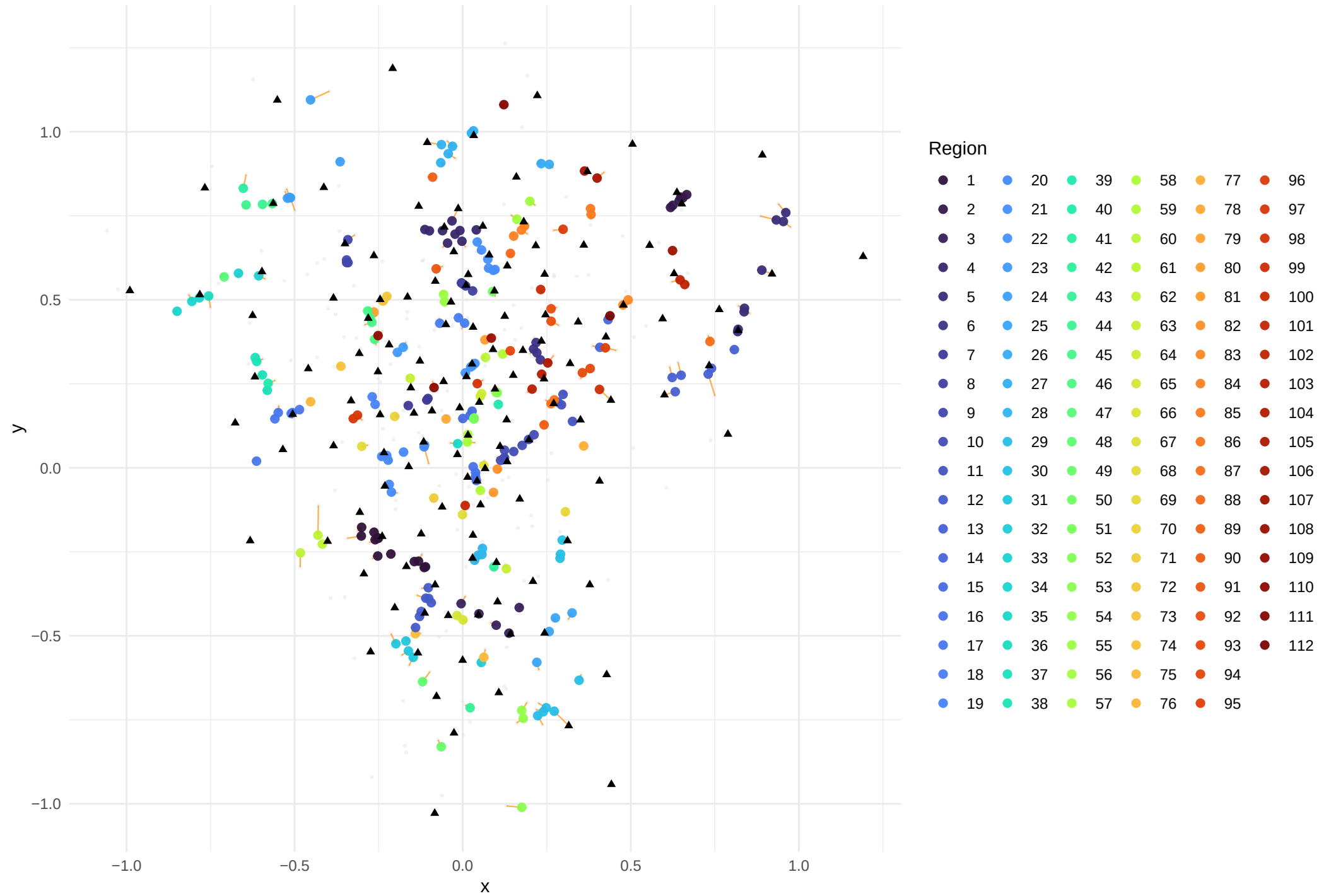
Intersection Regions (2+ anchors)

Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.



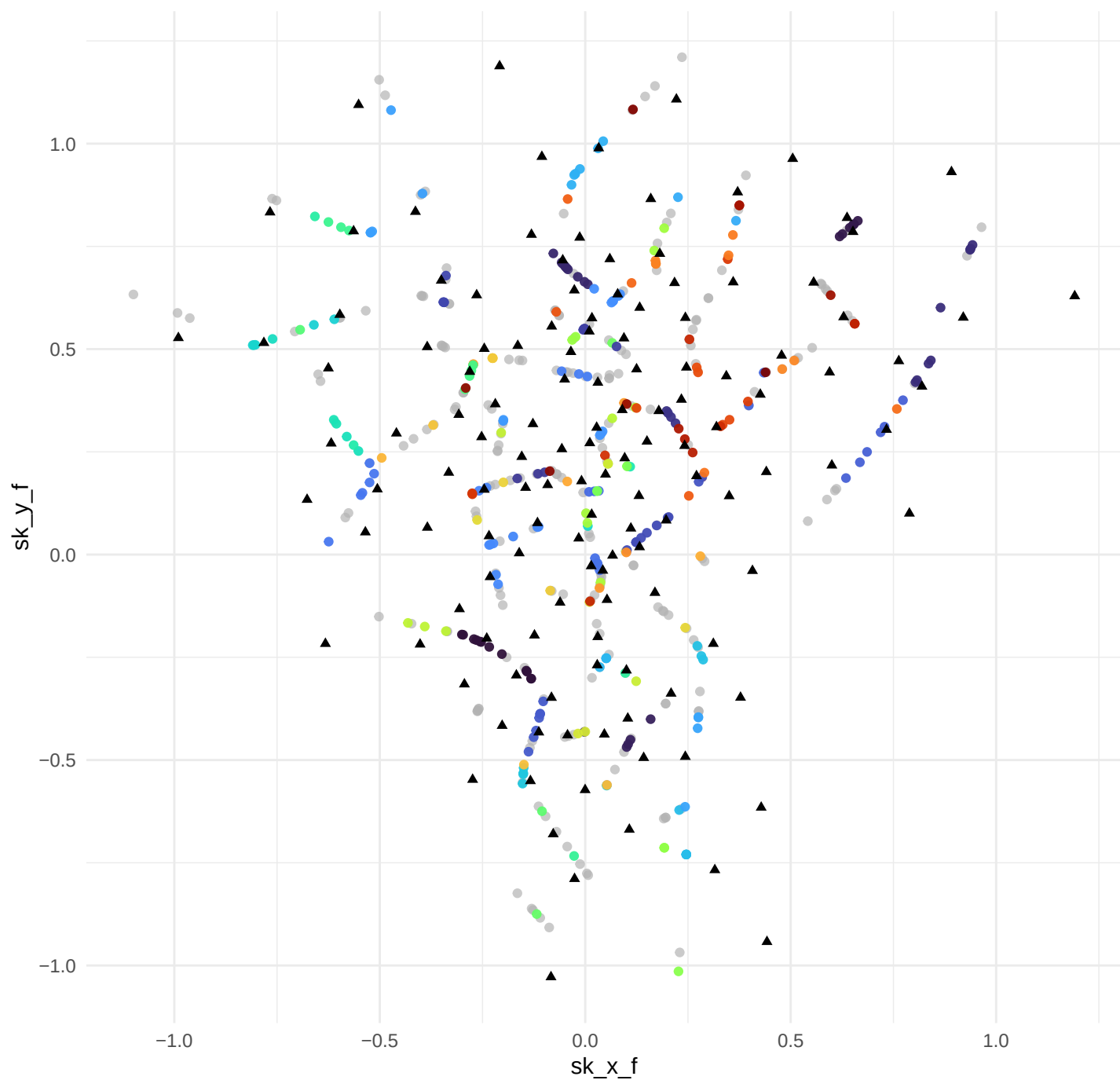
Parameters: k\_min = 5, g\_threshold = 3.0, o\_max = 0.3, o\_min = 0.10, stability = 0.90, scale\_factor = 2.5, max\_iter = 50, conv\_tol = 0.01

Stitched Backbone (Iteration 1)  
Orange path: stitching; Triangles: original anchors



# Final Stitched Backbone (Converged)

Final stitched points

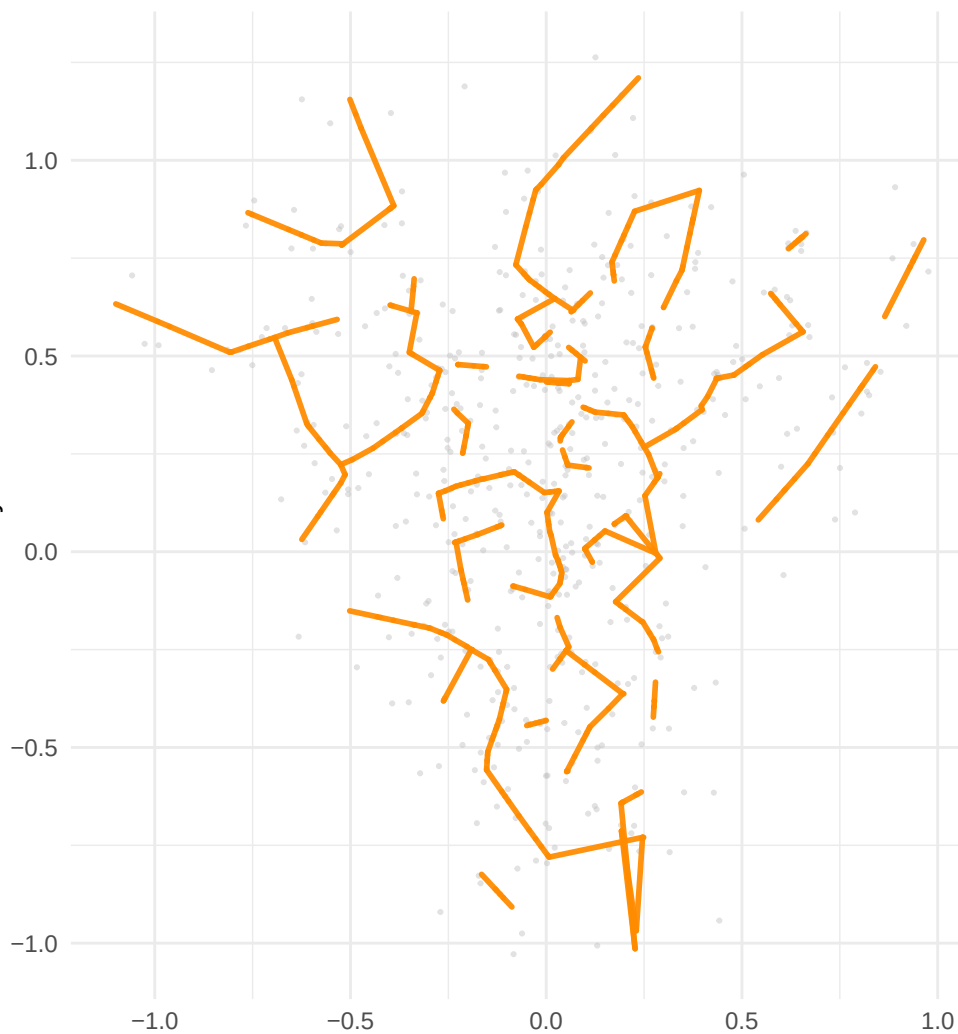


## Region

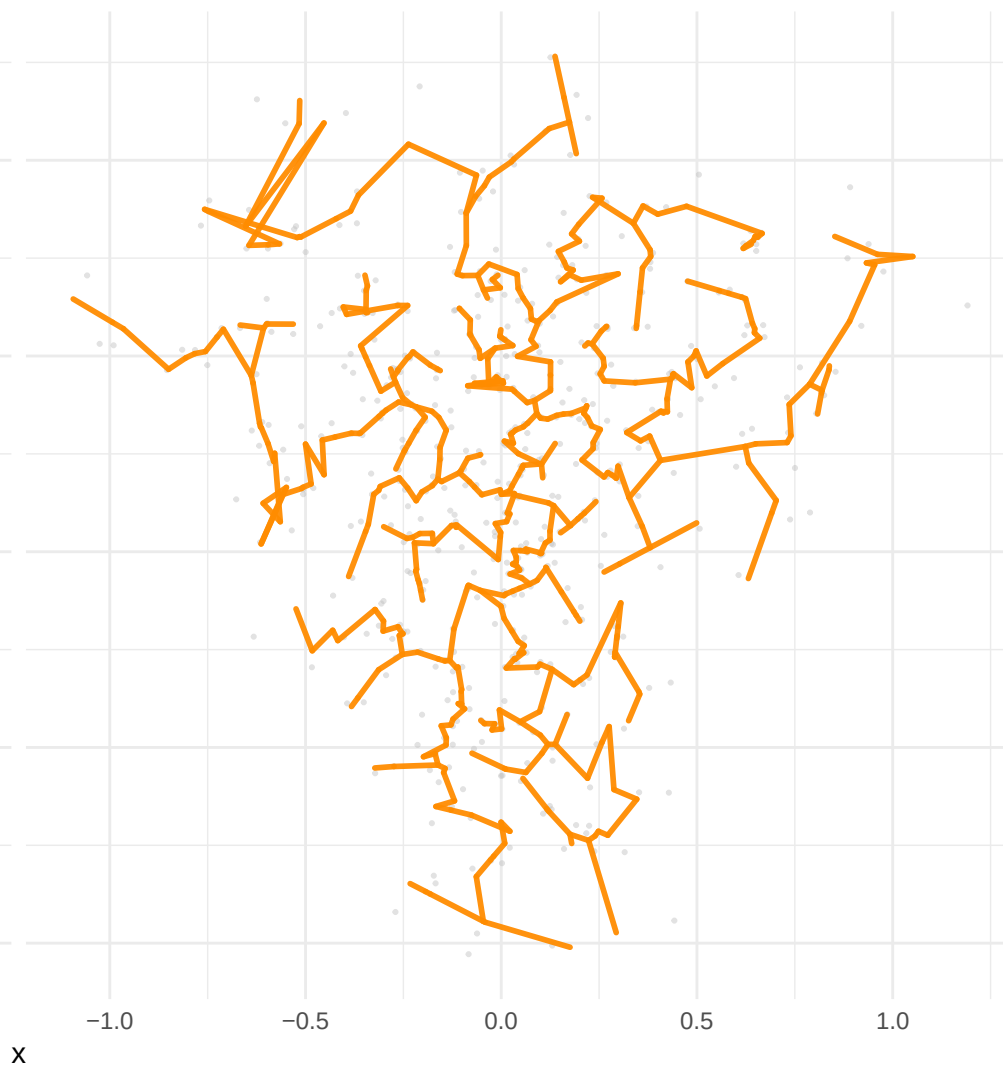
|    |    |    |    |    |     |
|----|----|----|----|----|-----|
| 1  | 20 | 39 | 58 | 77 | 96  |
| 2  | 21 | 40 | 59 | 78 | 97  |
| 3  | 22 | 41 | 60 | 79 | 98  |
| 4  | 23 | 42 | 61 | 80 | 99  |
| 5  | 24 | 43 | 62 | 81 | 100 |
| 6  | 25 | 44 | 63 | 82 | 101 |
| 7  | 26 | 45 | 64 | 83 | 102 |
| 8  | 27 | 46 | 65 | 84 | 103 |
| 9  | 28 | 47 | 66 | 85 | 104 |
| 10 | 29 | 48 | 67 | 86 | 105 |
| 11 | 30 | 49 | 68 | 87 | 106 |
| 12 | 31 | 50 | 69 | 88 | 107 |
| 13 | 32 | 51 | 70 | 89 | 108 |
| 14 | 33 | 52 | 71 | 90 | 109 |
| 15 | 34 | 53 | 72 | 91 | 110 |
| 16 | 35 | 54 | 73 | 92 | 111 |
| 17 | 36 | 55 | 74 | 93 | 112 |
| 18 | 37 | 56 | 75 | 94 |     |
| 19 | 38 | 57 | 76 | 95 |     |

## Backbone Edges

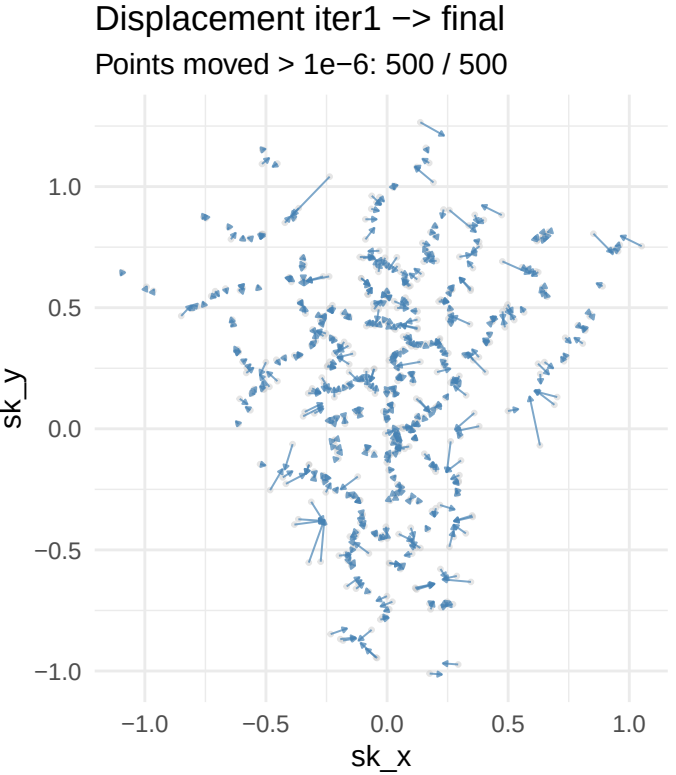
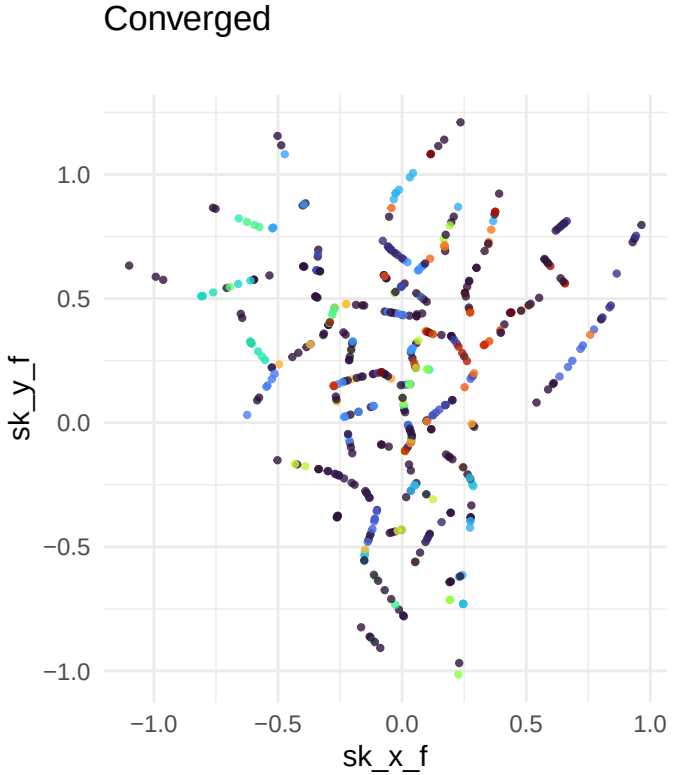
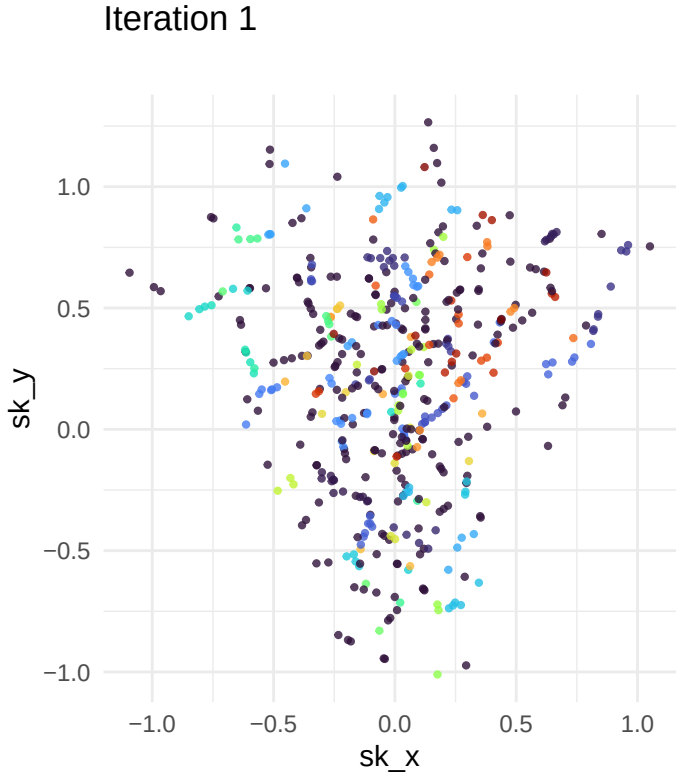
Final



Iteration 1



Convergence: iteration 1 vs final



Parameters: k\_min = 5, g\_threshold = 3.0, o\_max = 0.3, o\_min = 0.10, stability = 0.90, scale\_factor = 2.5, max\_iter = 50, conv\_tol = 0.01